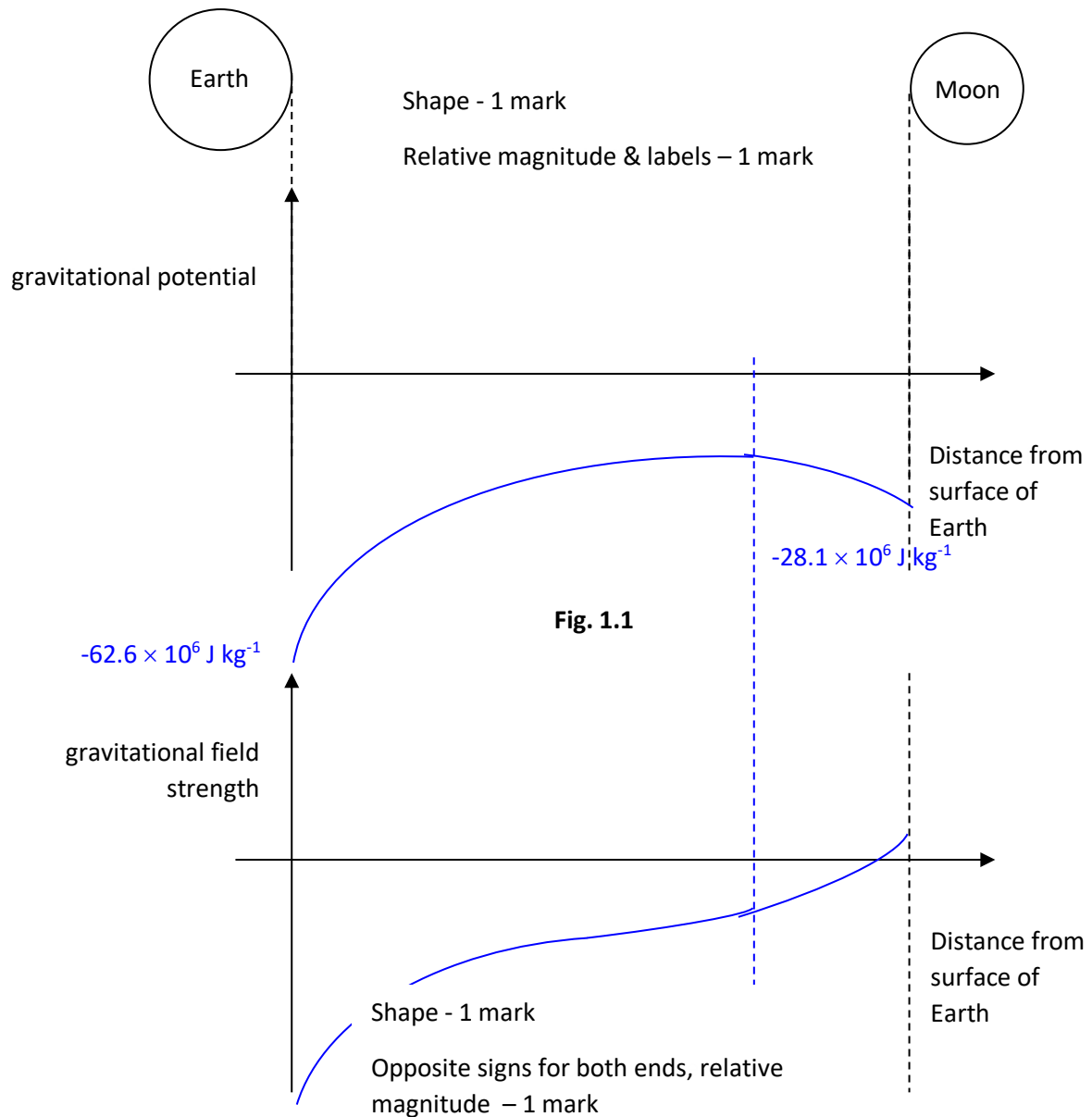


1 (a) The gravitational potential at a point within a gravitational field is the work done per unit mass by an external agent in bringing a body from infinity to that point (without any change in kinetic energy). B1



(b) (i) $g = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11})(6.4 \times 10^{23})}{(3.4 \times 10^6)^2} = 3.69 \text{ N kg}^{-1} = 3.7 \text{ N kg}^{-1}$ M1, A0

- (i) Since the height of 1800 m is much smaller than the radius of the planet, it can be assumed that the stone is moved through 1800 m in a uniform gravitational field.

Hence $\Delta E_p = mg\Delta h = 2.4(3.7)(1800) = 1.6 \times 10^4 \text{ J}$ M1, A1

- (ii) Assumption: All the loss in gravitational potential energy of the rock converted to kinetic energy.

Total final KE + GPE = Total initial KE + GPE

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{8r}\right) = 0 + 0$$

$$\Rightarrow v = \sqrt{\frac{GM}{4r}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.4 \times 10^{23})}{4(3.4 \times 10^6)}} = 1.77 \times 10^3 \text{ m s}^{-1}$$

M1, A1